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SEMICONDUCTOR DEVICE

Abstract

A semiconductor device may include a substrate including an active region extending in a first direction; a device isolation layer defining the active region; first and second gate electrodes extending in a second direction and spaced apart from each other; first channel layers apart from each other and surrounded by the first gate electrode; second channel layers apart from each other and surrounded by the second gate electrode; an insulating isolation pattern between the first and second gate electrodes and between the first and second channel layers; gate dielectric layers between the first channel layers and the first gate electrode and between the second channel layers and the second gate electrode; and epitaxial layers on opposing side surfaces of the first and second channel layers. Portions of the epitaxial layers may overlap the insulating isolation pattern, gate dielectric layers, and first and second gate electrodes in a third direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims benefit of priority to Korean Patent Application No. 10-2024-0033687 filed on Mar. 11, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Example embodiments of the present disclosure relate to a semiconductor device.

[0003] As demand for high performance, high speed, and/or multifunctionality for a semiconductor device has increased, integration density of a semiconductor devices has increased. To overcome limitations in operation properties caused by reduction of a size of a planar MOSFET (metal oxide semiconductor FET), there has been an effort for developing a semiconductor device including a FinFET including a fin-shaped channel and a gate-all-around field-effect transistor including nanosheets surrounded by a gate.

SUMMARY

[0004] An example embodiment of the present disclosure is to provide a semiconductor device having improved electrical properties and reliability.

[0005] According to an example embodiment, a semiconductor device may include a substrate including a first region and a second region, the substrate including an active region extending in a first direction; a device isolation layer defining the active region, the device isolation layer being on the substrate; a first gate electrode and a second gate electrode extending in a second direction, the second direction intersecting the active region, the first gate electrode and the second gate electrode being spaced apart from each other; a plurality of first channel layers spaced apart from each other in a third direction, the third direction being perpendicular to an upper surface of the substrate, and the plurality of first channel layers being surrounded by the first gate electrode on the first region; a plurality of second channel layers spaced apart from each other in the third direction, the plurality of second channel layers being surrounded by the second gate electrode on the second region; an insulating isolation pattern extending in the first direction between the first gate electrode and the second gate electrode and between the plurality of first channel layers and the plurality of second channel layers, the insulating isolation pattern being on the substrate; gate dielectric layers between the plurality of first channel layers and the first gate electrode, and the gate dielectric layers between the plurality of second channel layers and the second gate electrode; and epitaxial layers on side surfaces of the plurality of first channel layers and the plurality of second channel layers that respectively oppose each other. A portion of each of the epitaxial layers may overlap the insulating isolation pattern, the gate dielectric layers, the first gate electrode, and the second gate electrode in the third direction.

[0006] According to an example embodiment, a semiconductor device may include a substrate including an active region extending in a first direction; a first gate electrode and a second gate electrode extending in a second direction, the second direction intersecting the active region, the first gate electrode and the second gate electrode being spaced apart from each other; a plurality of first channel layers spaced apart from each other in a third direction, the third direction

perpendicular to an upper surface of the substrate, and the plurality of first channel layers being surrounded by the first gate electrode; a plurality of second channel layers spaced apart from each other in the third direction, the plurality of second channel layers being surrounded by the second gate electrode; an insulating isolation pattern between the plurality of first channel layers and the plurality of second channel layers, the insulating isolation pattern extending to a level lower than a level of an upper surface of the active region; first epitaxial layers on side surfaces of the plurality of first channel layers and the plurality of second channel layers, opposing each other, the first epitaxial layers being in contact with the insulating isolation pattern in the active region; and a second epitaxial layer in the active region of the substrate and having an upper surface in contact with the insulating isolation pattern.

[0007] According to an example embodiment, a semiconductor device may include a substrate including a first region and a second region, the substrate including an active region extending in a first direction; a first gate electrode and a second gate electrode extending in a second direction, the second direction intersecting the active region, the first gate electrode including a first conductive layer, the second gate electrode including a second conductive layer, the first gate electrode and the second gate electrode each including a third conductive layer, the first conductive layer and the third conductive layer of the first gate electrode being stacked in order, and the second conductive layer and the third conductive layer of the second gate electrode being stacked in order; a plurality of channel layers stacked in a third direction on the active region, the third direction being perpendicular to an upper surface of the substrate, and the plurality of channel layers overlapping the third conductive layer of the first gate electrode in the third direction; a gate dielectric layer covering a portion of each of the plurality of channel layers; an insulating isolation pattern between the first gate electrode and the second gate electrode; and a semiconductor material layer disposed on a side surface of at least one of the plurality of channel layers, wherein the first conductive layer and the second conductive layer may include different metal materials.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] The and other aspects, features, and advantages in example embodiments will be more clearly understood from the following detailed description, taken in combination with the accompanying drawings, in which:

[0009] FIG. 1 is a plan diagram illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0010] FIG. 2A is a cross-sectional diagram illustrating a semiconductor device taken along line I-I' according to example embodiments of the present disclosure;

[0011] FIG. 2B is a cross-sectional diagram illustrating a semiconductor device taken along line II-II' according to an example embodiment of the present disclosure;

[0012] FIG. 2C is a cross-sectional diagram illustrating a semiconductor device taken along line III-III' according to an example embodiment of the present disclosure;

[0013] FIG. 3 is a cross-sectional diagram illustrating a semiconductor device according to example embodiments of the present disclosure;

[0014] FIG. 4 is a cross-sectional diagram illustrating a semiconductor device according to example embodiments of the present disclosure;

[0015] FIG. 5 is a cross-sectional diagram illustrating a semiconductor device according to example embodiments of the present disclosure;

[0016] FIG. 6 is a cross-sectional diagram illustrating a semiconductor device according to example embodiments of the present disclosure;

[0017] FIG. 7 is a cross-sectional diagram illustrating a semiconductor device according to

example embodiments of the present disclosure; and
[0018] FIGS. **8A** to **8I** are diagrams illustrating processes of a method of manufacturing a semiconductor device according to example embodiments of the present disclosure.

DETAILED DESCRIPTION

[0019] Hereinafter, embodiments in the example embodiment will be described as follows with reference to the accompanying drawings.

[0020] FIG. **1** is a plan diagram illustrating a semiconductor device **100** according to an example embodiment.

[0021] FIGS. **2A** to **2C** are cross-sectional diagrams illustrating the semiconductor device **100** according to example embodiment. FIG. **2A** is a cross-sectional diagram illustrating the semiconductor device **100** taken along line I-I' according to example embodiments; FIG. **2B** is a cross-sectional diagram illustrating the semiconductor device **100** taken along line II-II' according to an example embodiment; FIG. **2C** is a cross-sectional diagram illustrating the semiconductor device **100** taken along line III-III' according to an example embodiment.

[0022] Referring to FIGS. **1** to **2C**, a semiconductor device **100** may include a substrate **101** having first and second regions **R1** and **R2**, an active region **105** on the substrate **101**, channel structures **140** including a plurality of channel layers **141**, **142**, and **143** vertically disposed and spaced apart from each other on the active region **105**, a first gate electrode **170A** and a second gate electrode **170B** extending by intersecting the active region **105**, an insulating isolation pattern **130** isolating the first gate electrode **170A** from the second gate electrode **170B**, semiconductor material layers **EP**, source/drain regions **150** in contact with the channel structures **140**, and contact plugs **180** connected to the source/drain regions **150**. The semiconductor device **100** may further include a device isolation layer **110**, gate dielectric layers **162**, gate spacer layers **163**, gate capping layers **166**, and first and second interlayer insulating layers **IL1** and **IL2**.

[0023] In the semiconductor device **100**, the active region **105** may have a fin structure, the first and second gate electrodes **170A** and **170B** may be disposed between the active region **105** and the channel structure **140**, between the plurality of channel layers **141**, **142**, and **143** in the channel structure **140**, and on the channel structure **140**, and an insulating isolation pattern **130** may be disposed between the plurality of first and second channel layers spaced apart from each other in the horizontal direction. Accordingly, the semiconductor device **100** may include transistors with a multi-bridge channel FET (MBCFET™) structure, which is a gate-all-around (Gate-All-Around) type field effect transistor, or transistors having a forksheet structure including insulating pillars between a plurality of nanosheets.

[0024] The substrate **101** may have an upper surface extending in a first direction (e.g., X-axis direction) and a second direction (e.g., Y-axis direction). The substrate **101** may include a semiconductor material, such as a group IV semiconductor, a group III-V compound semiconductor, or a group II-VI compound semiconductor. For example, a group IV semiconductor may include silicon, germanium, or silicon-germanium. The substrate **101** may be provided as a bulk wafer, an epitaxial layer, a silicon on insulator (SOI) layer, or a semiconductor on insulator (SeOI) layer.

[0025] The substrate **101** may include first region **R1** and second region **R2**, and the first and second regions **R1** and **R2** may be adjacent to each other. A portion of the active region **105**, the source/drain regions **150**, and the first gate electrode **170A** may be disposed in the first region **R1**, and the other portion of the active region **105**, the source/drain regions **150**, and the second gate electrode **170B** may be disposed in the second region **R2**. For example, in the first region **R1**, nFET (n-type Field Effect Transistor) may be disposed, and in the second region **R2**, pFET (p-type Field Effect Transistor) may be disposed. In some example embodiments, in the first and second regions **R1** and **R2**, transistors having the same conductivity type and different electrical properties may be disposed.

[0026] The active region **105** may be defined by the device isolation layer **110** and may be disposed

to extend in the first direction (e.g., X-axis direction). Depending on descriptions, the active region **105** may be described as a portion of the substrate **101**. The active regions **105** may partially protrude to the device isolation layer **110**, such that upper surfaces of the active region **105** may be disposed on a level higher than a level of an upper surface of the device isolation layer **110**. The active regions **105** may include a portion of the substrate **101** or may include an epitaxial layer grown from the substrate **101**. However, the active region **105** may be partially recessed to form recess regions on both sides of the first and second gate electrodes **170A** and **170B**, and the source/drain regions **150** having different conductivity types may be disposed in the recess regions. [0027] In example embodiments, the region overlapping the first region R1 of the active region **105** may be referred to as a first active region, and the region overlapping the second region R2 of the active region **105** may be referred to as a second active region. Each of the first and second active regions may include well regions including impurities. For example, in the first active region of the first region R1 in which the nFET is disposed, the well region may include P-type impurities such as boron (B), gallium (Ga), or aluminum (Al). In the second active region of second region R2 in which the pFET is disposed, the well region may include N-type impurities such as phosphorus (P), arsenic (As), or antimony (Sb). For example, the well region may be disposed at a predetermined depth from an upper surface of each of the first and second active regions.

[0028] The device isolation layer **110** may define the active region **105** in the substrate **101**. The device isolation layer **110** may be formed, for example, by a shallow trench isolation (STI) process. The device isolation layer **110** may expose the upper surface of active region **105** and may partially expose an upper portion. In some example embodiments, the device isolation layer **110** may have a curved upper surface to have a level increasing toward the active region **105**. The device isolation layer **110** may be formed of an insulating material. The device isolation layer **110** may include, for example, oxide, nitride, or a combination thereof.

[0029] The channel structures **140** may include a plurality of first channel layers **140A** spaced apart from each other in the third direction (e.g., Z-axis direction) perpendicular to an upper surface of substrate **101**, and a plurality of second channel layers **140B** spaced apart from each other in the third direction (e.g., Z-axis direction) perpendicular to the upper surface of the substrate **101** on the second region R2. The plurality of first channel layers **140A** and the plurality of second channel layers **140B** may be spaced apart from each other in the second direction (e.g., Y-axis direction). The channel structures **140** may be connected to the source/drain regions **150** (e.g., **150A**, **150B**) and may be spaced apart from the upper surface of the active region **105**.

[0030] The plurality of channel layers **141**, **142**, and **143** may be formed of a semiconductor material, and may include, for example, at least one of silicon (Si), silicon germanium (SiGe), and germanium (Ge). For example, the plurality of channel layers **141**, **142**, and **143** may be formed of the same material as that of the substrate **101**. In example embodiments, the plurality of channel layers **141**, **142**, and **143** may include an impurity region disposed in a region adjacent to the source/drain regions **150**. The number of the channel layers **141**, **142**, and **143** of each of the channel structures **140** and the shape thereof may be varied in example embodiments.

[0031] The first source/drain region **150A** and the second source/drain region **150B** may be disposed on the first and second active regions on both sides of the first and second gate electrodes **170A** and **170B** and the channel structures **140**, respectively. The first source/drain regions **150A** may be disposed on the first active regions on both sides of the first gate electrode **170A**, and the second source/drain regions **150B** may be disposed on the second active regions on both sides of the second gate electrode **170B**.

[0032] The first and second source/drain regions **150A** and **150B** may be disposed in recess regions partially recessed into an upper portion of the first and second active regions. The first and second source/drain regions **150A** and **150B** may be in contact with the first to third channel layers **141**, **142**, and **143** of the channel structures **140**, and may be disposed to cover side surfaces of each of the plurality of channel layers **141**, **142**, and **143**. Upper surfaces of the first and second

source/drain regions **150A** and **150B** may be disposed on a level the same as or similar to a level of lower surfaces of the uppermost regions of the first and second gate electrodes **170A** and **170B**, and the level may be varied in example embodiments. In example embodiments, the first and second source/drain regions **150A** and **150B** may be connected or merged with each other on two or more first and second active regions adjacent to each other in the Y-direction, and may form the first and second source/drain regions **150A** and **150B**, respectively.

[0033] The first and second source/drain regions **150A** and **150B** may include impurities of different conductivity types. For example, the first source/drain regions **150A** may include N-type impurities, and the second source/drain regions **150B** may include P-type impurities, but an example embodiment thereof is not limited thereto.

[0034] The gate structure may include first and second gate electrodes **170A** and **170B**, a gate dielectric layer **162**, gate spacer layers **163**, and a gate capping layer **166**.

[0035] The first gate electrode **170A** and the second gate electrode **170B** may intersect the active region **105** and the channel structures **140** and may extend in the second direction (e.g., Y-axis direction) on the active region **105** and the channel structures **140**. Physical channel regions of transistors may be formed in the active region **105** and/or the channel structures **140** intersecting the first and second gate electrodes **170A** and **170B**. The first and second gate electrodes **170A** and **170B** may fill regions between the plurality of first channel layers **140A** and the second channel layers **140B**, respectively, on the active regions **105** and may extend to the channel structures **140**. The first and second gate electrodes **170A** and **170B** may be spaced apart from the first to third channel layers **141**, **142**, and **143** by the gate dielectric layer **162**.

[0036] The first and second gate electrodes **170A** and **170B** may be disposed linearly in the second direction (e.g., Y-axis direction). The first gate electrode **170A** and the second gate electrode **170B** may be disposed on the first and second regions R1 and R2, respectively, with the insulating isolation pattern **130** therebetween. The first gate electrode **170A** may include first and third conductive layers **171** and **173** stacked in order, and the second gate electrode **170B** may include second and third conductive layers **172** and **173** stacked in order. The first and second gate electrodes **170A** and **170B** may be included in nFET and pFET, respectively. Alternatively, the first and second gate electrodes **170A** and **170B** may be included in nFETs having different operating voltages, or pFETs having different operating voltages.

[0037] The first conductive layer **171** may be disposed on the gate dielectric layers **162** on the first region R1. The first conductive layer **171** may extend on the gate dielectric layers **162** extending conformally along a region between the plurality of first channel layers **140A**, and may extend along an upper surface of the device isolation layer **110** and a side surface of insulating isolation pattern **130**. The first conductive layer **171** may conformally extend along an upper surface, a side surface, and a lower surface of each of the plurality of first channel layers **140A**. The third conductive layer **173** may be disposed on the first conductive layer **171** and may fill a space in the first conductive layer **171** in a cross-sectional surface. The first conductive layer **171** may have substantially the same thickness in regions other than the plurality of first channel layers **140A**.

[0038] The notion that elements are “substantially the same” may indicate that the element may be completely the same and may also indicate that the elements may be determined to be the same in consideration of errors or deviations occurring during a process.

[0039] The second conductive layer **172** may be disposed on the gate dielectric layers **162** on the second region R2. The second conductive layer **172** may extend on the gate dielectric layers **162** extending conformally along a region between the plurality of second channel layers **140B**, and may extend along an upper surface of the device isolation layer **110** and a side surface of the insulating isolation pattern **130**. The second conductive layer **172** may conformally extend along an upper surface, a side surface, and a lower surface of each of the plurality of second channel layers **140B**. The third conductive layer **173** may be disposed on the second conductive layer **172** and may fill a space in the second conductive layer **172** on a cross-sectional surface. The second

conductive layer **172** may have substantially the same thickness in regions other than the plurality of second channel layers **140B**.

[0040] The first and second gate electrodes **170A** and **170B** may be electrically isolated from each other by the insulating isolation pattern **130**. The first and second gate electrodes **170A** and **170B** may have an internal stack structure symmetrical to each other with respect to the insulating isolation pattern **130**, but an example embodiment thereof is not limited thereto.

[0041] The first and second gate electrodes **170A** and **170B** may include a conductive material, for example, a metal nitride such as titanium nitride (TiN), tantalum nitride (TaN), or tungsten nitride (WN), and/or may include a metal material such as aluminum (Al), tungsten (W), or molybdenum (Mo), or a semiconductor material such as doped polysilicon. The first conductive layer **171** may include a material different from that of the second and third conductive layers **172** and **173**, and the second and third conductive layers **172** and **173** may include the same material or different materials. Even when the second and third conductive layers **172** and **173** may include the same material, the second and third conductive layers **172** and **173** may be formed in different processes, and an interfacial surface therebetween may be distinct.

[0042] For example, the first to third conductive layers **171**, **172**, and **173** may be metal layers. In an example embodiment, the second conductive layer **172** may have a work function equal to or smaller than that of the first conductive layer **171**. For example, the first and second conductive layers **171** and **172** may be configured as metal layers for controlling the work function, and the first conductive layer **171** may include P-type metal and the second conductive layer **172** may include N-type metal. For example, the first conductive layer **171** may include at least one of TiAlN and TiN, and the second conductive layer **172** may include at least one of TiAlC and TiN, but an example embodiment thereof is not limited thereto.

[0043] The gate dielectric layers **162** may be disposed between the active region **105** and the first and second gate electrodes **170A** and **170B**, between the plurality of first channel layers **140A** and the first gate electrode **170A**, and between the plurality of second channel layers **140B** and the second gate electrode **170B**, and may be disposed to cover at least a portion of surfaces of the first and second gate electrodes **170A** and **170B**. For example, the gate dielectric layers **162** may be disposed to surround the entirety of surfaces other than upper surfaces of the first and second gate electrodes **170A** and **170B**. The gate dielectric layers **162** may extend to a region between the first and second gate electrodes **170A** and **170B** and the gate spacer layers **163**, but an example embodiment thereof is not limited thereto.

[0044] The gate dielectric layers **162** may include oxide, nitride, or high-K material. The high-K material may refer to a dielectric material having a dielectric constant higher than that of a silicon oxide film ((SiO.sub.2)). The high-K material may include, for example, at least one of aluminum oxide (Al.sub.2O.sub.3), tantalum oxide (Ta.sub.2O.sub.3), titanium oxide (TiO.sub.2), yttrium oxide (Y.sub.2O.sub.3), zirconium oxide (ZrO.sub.2), zirconium silicon oxide (ZrSi.sub.xO.sub.y), hafnium oxide (HfO.sub.2), hafnium silicon oxide (HfSi.sub.xO.sub.y), lanthanum oxide (La.sub.2O.sub.3), lanthanum aluminum oxide (LaAl.sub.xO.sub.y), lanthanum hafnium oxide (LaHf.sub.xO.sub.y), hafnium aluminum oxide (HfAl.sub.xO.sub.y), and praseodymium oxide (Pr.sub.2O.sub.3). In example embodiments, the gate dielectric layer **162** may include multiple films.

[0045] The gate spacer layers **163** may be disposed on both side surfaces of the first and second gate electrodes **170A** and **170B**, respectively. The first and second gate electrodes **170A** and **170B** may insulate the first and second source/drain regions **150A** and **150B** from the first and second gate electrodes **170A** and **170B**. The gate spacer layers **163** may be stacked and may form a multilayer structure. However, in some example embodiments, the gate spacer layer may include a single layer. The gate spacer layers **163** may include at least one of oxide, nitride, and oxynitride and, for example, may include a low dielectric constant film.

[0046] The gate capping layers **166** may be disposed on upper surfaces of the first and second gate

electrodes **170A** and **170B** and an upper surface of the insulating isolation pattern **130**. The gate capping layers **166** may include at least one of oxide, nitride, and oxynitride.

[0047] The insulating isolation pattern **130** may be disposed between the first gate electrode **170A** and the second gate electrode **170B** to isolate the first and second gate electrodes **170A** and **170B** from each other. An upper surface of the insulating isolation pattern **130** may be coplanar with upper surfaces of the first and second gate electrodes **170A** and **170B**. An upper surface of the insulating isolation pattern **130** may be exposed from the gate dielectric layers **162**, the first conductive layer **171** and the second conductive layer **172** and may be covered by the gate capping layer **166**. In some example embodiments, an upper surface of the insulating isolation pattern **130** may be covered with another layer formed of an insulating material, such as an interlayer insulating layer. Side surfaces of the insulating isolation pattern **130** may be perpendicular to an upper surface of the substrate **101** or may be inclined. In an example embodiment, the insulating isolation pattern **130** may have a shape tapered toward the substrate **101**.

[0048] At least a portion of a side surface **130CS** of the insulating isolation pattern **130**, in the second direction (e.g., Y-axis direction), may have an inwardly concave shape on a level between channel layers **140** spaced apart from each other in the vertical direction. At least a portion of a side surface in the second direction of the insulating isolation pattern **130** may be in contact with the first epitaxial layer **145**. The first epitaxial layer **145** may have a convex shape toward the insulating isolation pattern **130**, and a side surface of the insulating isolation pattern **130** in the second direction may have a convex shape or a concave shape depending on a shape of the adjacent first epitaxial layer **145**. The side surface of the insulating isolation pattern **130** in the second direction may have a serrated shape in which a convex shape and a concave shape are repeatedly formed. The gate dielectric layers **162**, the first conductive layer **171** (or the second conductive layer **172**) and the third conductive layer **173** may be stacked in order on the concave side surface of the insulating isolation pattern **130**.

[0049] A shape of a lower surface of the insulating isolation pattern **130** may be varied depending on a shape of the adjacent second epitaxial layer **146**. A lower surface of the insulating isolation pattern **130** may have a shape inclined toward the substrate **101** in a direction away from the first and second gate electrodes **170A** and **170B**. Among the lower surfaces of the insulating isolation pattern **130**, a lower surface of a central portion adjacent to a boundary between the first region **R1** and the second region **R2** may have a convex shape toward the substrate **101**. A lower surface of a central portion of the insulating isolation pattern **130** may have a shape surrounded by the second epitaxial layer **146**. A lowermost end of the insulating isolation pattern **130** may be disposed on a level lower than a level of a lowermost end of the first and second gate electrodes **170A** and **170B**, and may be disposed on a level higher than a level of an uppermost end of the substrate **101**.

[0050] A width of the insulating isolation pattern **130** in the second direction (e.g., Y-axis direction) between ends **145e** of the first epitaxial layers **145** opposing each other may be referred to as a first width **W1**. A width of the portion in the second direction surrounded by the second epitaxial layer **146** of the insulating isolation pattern **130** may be referred to as a second width **W2**. A portion of the insulating isolation pattern **130**, surrounded by the second epitaxial layer **146**, may refer to a convex portion toward the substrate **101**. Sizes of the first width **W1** and the second width **W2** may be substantially the same. Here, the notion that the sizes are the same may include process errors, and may indicate that a width may not be intentionally designed differently.

[0051] A width between the side surfaces **130CS** of the insulating isolation pattern **130** opposing each other on a level between the channel layers **140**, spaced apart from each other in the vertical direction, may be referred to as a third width **W3**. A width of the insulating isolation pattern **130** on a level higher than a level of the uppermost channel layer **143** among the plurality of channel layers **140** may be referred to as a fourth width **W4**. A size of the third width **W3** may be substantially the same as a size of the first width **W1**. A size of the fourth width **W4** may be the same as or greater than a size of the first width **W1**. As a level of the insulating isolation pattern **130** in the third

direction (e.g., Z-axis direction) changes, changes in the size of the width may be repeated.

[0052] The insulating isolation pattern **130** may include an insulating material. The insulating isolation pattern **130** may include, for example, at least one of silicon oxide, silicon nitride, silicon oxynitride, and silicon carbide.

[0053] The semiconductor material layers EP may include first epitaxial layers **145** disposed on side surfaces of the plurality of first and second channel layers **140A** and **140B** opposing each other, and a second epitaxial layer **146** disposed in the active region **105**.

[0054] The first epitaxial layers **145** may be disposed on side surfaces **140S** on which the plurality of channel layers **140** oppose each other. Among the first epitaxial layers **145**, the first epitaxial layer **145** on a side surface of the first channel layer **140A** and the first epitaxial layer **145** on a side surface of the second channel layer **140B** disposed on the same level may be spaced apart from each other by the insulating isolation pattern **130**. other. A width of each of the first epitaxial layers **145** in the third direction (e.g., Z-axis direction) may decrease toward the insulating isolation pattern **130**. In an example embodiment, a cross-sectional surface of each of the first epitaxial layers **145** may have a triangular shape of which a width in the third direction may decrease toward the insulating isolation pattern **130**, but an example embodiment thereof is not limited thereto. In an example embodiment, the cross-sectional surface of the first epitaxial layers **145** may have a triangular shape of which ends **145e** opposing each other correspond to apexes.

[0055] The second epitaxial layer **146** may be disposed on the substrate **101**. A width of the second epitaxial layer **146** in the second direction (e.g., the Y-axis direction) may be substantially the same as a width from an interfacial surface between the first channel layers **140A** and the first epitaxial layer **145** to an interfacial surface between the second channel layers **140B** and the first epitaxial layer **145**. Here, the notion that the sizes are the same may include process errors, and may indicate that a width is not intentionally designed differently. An upper surface of the second epitaxial layer **146** may have a partially inclined shape. Among upper surfaces of the second epitaxial layer **146**, an upper surface of a central portion adjacent to a boundary between the first region R1 and the second region R2 may have a concave shape toward the substrate **101**. An upper surface of a central portion of the second epitaxial layer **146** may have a shape surrounding the insulating isolation pattern **130**. An upper surface of the second epitaxial layer **146** may be in contact with the insulating isolation pattern **130**.

[0056] The first epitaxial layer **145** and the second epitaxial layer **146** may include the same material, and in an example embodiment, the first epitaxial layer **145** and the second epitaxial layer **146** may include selectively epitaxially grown silicon (Si). In an example embodiment, the first epitaxial layer **145** may include the same material as that of the plurality of channel layers **140**, and a boundary of the surface on which the first epitaxial layer **145** and the channel layer **140** meet may not be distinct. In example embodiments, the first epitaxial layer **145** may correspond to a protrusion formed on the plurality of channel layers.

[0057] At least a portion of each of the first epitaxial layers **145** may overlap the insulating isolation pattern **130**, the gate dielectric layers **162**, the first and second gate electrodes **170A** and **170B** in the third direction (e.g., Z-axis direction), respectively. Among the plurality of first channel layers **140A**, the third conductive layer **173** may overlap at least a portion of the first epitaxial layer **145** in the third direction. The semiconductor device in the example embodiment, by forming a first epitaxial layer **145** extending from a side surface of the plurality of channel layers **140** and including the first and second gate electrodes **170A** and **170B** extending inwardly of the insulating isolation pattern **130**, an effective area and a length of the channel layers **140** may be lengthened, such that electrical signal transmission may be smoothly performed, thereby improving electrical properties and reliability of the semiconductor device **100**.

[0058] The first interlayer insulating layer IL1 may cover source/drain regions **150**. The first interlayer insulating layer IL1 may include at least one of oxide, nitride, and oxynitride, and may include, for example, a low dielectric constant material. In example embodiments, the first

interlayer insulating layer **IL1** may include a plurality of insulating layers.

[0059] The contact plugs **180** may penetrate the first interlayer insulating layer **IL1**, may be connected to the source/drain regions **150**, and may apply an electrical signal to the source/drain regions **150**. The contact plugs **180** may have an inclined side surface on which a width of the lower portion may decrease further than a width of the upper portion depending on the aspect ratio, but an example embodiment thereof is not limited thereto. The contact plugs **180** may extend downwardly, for example, to a region below a lower surface of the channel layer **143** of an uppermost first channel layer of the plurality of first channel layers **140A**, but an example embodiment thereof is not limited thereto. In some example embodiments, the contact plugs **180** may not recess source/drain regions **150**, and may be in contact with upper surfaces of the source/drain regions **150**.

[0060] Each of the contact plugs **180** may include a metal-semiconductor compound layer, for example, a metal silicide layer, disposed on a lower end including a lower surface, and may further include a barrier layer forming side surfaces of the contact plug **180** and extending to an upper surface of the metal-semiconductor compound layer. For example, the barrier layer may include a metal nitride such as titanium nitride (TiN), tantalum nitride (TaN), or tungsten nitride (WN). The contact plugs **180** may include a metal material such as aluminum (Al), tungsten (W), or molybdenum (Mo). In example embodiments, the number of the conductive layers included in the contact plugs **180** and the arrangement form thereof may be varied.

[0061] An interconnection structure such as a contact plug may be further disposed on the first and second gate electrodes **170A** and **170B**, and an interconnection structure such as an interconnection line **M1** connected to the contact plugs **180** may be further disposed on the contact plugs **180**.

[0062] In the description in the example embodiments below, descriptions overlapping the descriptions described above with reference to FIGS. **1** to **2C** may not be provided.

[0063] FIG. **3** is a cross-sectional diagram illustrating a semiconductor device according to example embodiments, illustrating a region corresponding to FIG. **2B**.

[0064] Referring to FIG. **3**, a semiconductor device **100A** in an example embodiment may be configured the same as the example described with reference to FIGS. **1** to **2C**, other than the configuration in which a cross-sectional surface of the first epitaxial layer **145** has a trapezoid shape. The semiconductor device **100A** may include a first epitaxial layer **145** of which the cross-sectional surface has a trapezoidal shape. A width of the first epitaxial layer **145** of the semiconductor device **100A** in the second direction (e.g., Y-axis direction) may be smaller than a width of the first epitaxial layer **145** of the semiconductor device **100** (see FIG. **2B**) in the second direction. In the semiconductor device **100A** in the example embodiment, an end **145e** of the first epitaxial layer **145** may be substantially parallel to a side surface **140S** on which the channel layers **140** oppose each other. The first epitaxial layer **145** may have a shape of which a width in the third direction (e.g., the Z-axis direction) may decrease toward the insulating isolation pattern **130**. A width of the second epitaxial layer **146** of the semiconductor device **100A** in the second direction (e.g., the Y-axis direction) and the third direction (e.g., the Z-axis direction) from an interfacial surface with the active region **105** may be smaller than a width in the second and third directions from an interfacial surface from the active region **105** of the second epitaxial layer **146** of the semiconductor device (**100**, see FIG. **2B**). In example embodiments, a cross-sectional surface of each of the first epitaxial layers **145** may have a quadrangular shape. In an example embodiment, the first epitaxial layer **145** may have a rectangular cross-sectional surface shape of which a width may be constantly maintained in the third direction.

[0065] Referring to FIG. **3** together with FIG. **2B**, a side surface of the insulating isolation pattern **130** may have a serrated shape in which a convex shape and a concave shape are repeated along shapes of the first epitaxial layer **145**, the first gate electrode **170A** and the second gate electrode **170B**, in contact with each other. A first width **W1** of a portion of the insulating isolation pattern **130** between ends **145e** of the first epitaxial layers **145** opposing each other may be substantially

equal to a second width W2 of a portion of the insulating isolation pattern 130 surrounded by the second epitaxial layer 146. Here, the configuration in which the sizes are the same may include process errors and may indicate that the width is not intentionally designed differently. The first width W1 and second width W2 of the insulating isolation pattern 130 in the semiconductor device 100A may be greater than the first width W1 and the second width W2 of the insulating isolation pattern 130 in the semiconductor device (100, see FIG. 2B). The first width W1 and the second width W2 of the insulating isolation pattern 130 in the semiconductor device 100A may be greater than a fourth width W4. On a level between the channel layers 140 spaced apart from each other in the vertical direction, the third width W3 between the side surfaces 130CS of the insulating isolation pattern 130, opposing each other, may be smaller than the first width W1.

[0066] FIG. 4 is a cross-sectional diagram illustrating a semiconductor device according to example embodiments, illustrating a region corresponding to FIG. 2B.

[0067] Referring to FIG. 4, a semiconductor device 100B in an example embodiment may be configured the same as the example described with reference to FIGS. 1 to 3, other than the configuration in which one end 145e of the first epitaxial layer 145 has a rounded shape. The semiconductor device 100B may include a first epitaxial layer 145 of which a width in the third direction (e.g., the Z-axis direction) may decrease toward the insulating isolation pattern 130. In the semiconductor device 100B in the example embodiment, the end 145e of the first epitaxial layer 145 may have a circular shape or a rounded shape.

[0068] A first width W1 of the portion of the insulating isolation pattern 130 between the ends 145e of the first epitaxial layers 145, opposing each other, may be the same as or greater than a second width W2 of a portion of the insulating isolation pattern 130 surrounded by the second epitaxial layer 146. A fourth width W4 of the insulating isolation pattern 130 in the semiconductor device 100B may be substantially the same as or greater than the first width W1. The first width W1 and the second width W2 of the insulating isolation pattern 130 in the semiconductor device 100B may be greater than the first width W1 and the second width W2 of the insulating isolation pattern 130 in the semiconductor device 100 (see FIG. 2B). On a level between channel layers 140 spaced apart from each other in the vertical direction, the third width W3 between the side surfaces 130CS of the insulating isolation pattern 130, opposing each other, may be smaller than the first width W1.

[0069] FIG. 5 is a cross-sectional diagram illustrating a semiconductor device according to example embodiments, illustrating a region corresponding to FIG. 2B.

[0070] Referring to FIG. 5, a semiconductor device 100C in an example embodiment may be configured the same as or similar to the example described with reference to FIGS. 1 to 4, other than the configuration in which at least a portion 130LS of the side surface of the insulating isolation pattern 130 extends linearly along ends 145e of the first epitaxial layers 145 in the second direction. In the semiconductor device 100C, at least a portion 130LS of the side surface of the insulating isolation pattern 130 may be configured to extend in the third direction (e.g., Z-axis direction) along the ends 145e of the first epitaxial layers 145 in the second direction (e.g., Y side direction). The side surface 130LS of the insulating isolation pattern 130 may be substantially parallel to the side surfaces 140S of the plurality of channel layers 140, opposing each other.

[0071] The first width W1 of a portion of the insulating isolation pattern 130 between the ends 145e of the first epitaxial layers 145, opposing each other, may be substantially the second width W2 of the portion of the insulating isolation pattern 130 surrounded by the second epitaxial layer 146. On a level between the channel layers 140 spaced apart from each other in the vertical direction, a third width W3 between side surfaces 130CS of the insulating isolation pattern 130, opposing each other, may be substantially the same as the first width W1. A fourth width W4 of the insulating isolation pattern 130 in the semiconductor device 100C may be greater than the first width W1, the second width W2 and the third width W3. As another example, the fourth width W4 of the insulating isolation pattern 130 in the semiconductor device 100C may be substantially the same as the first width W1, the second width W2 and the third width W3. Here, the configuration

in which the sizes are the same may include process errors, and may indicate that the width is not intentionally designed differently. The insulating isolation pattern **130** may have a shape of which a width may be constant in the second direction (e.g., Y-axis direction) in a region from the ends **145e** of an uppermost first epitaxial layers **145** to the second epitaxial layer **146**.

[0072] The gate dielectric layers **162**, a first conductive layer **171** and a third conductive layer **173** may be stacked in order on at least a portion of the side surface **130LS** of the insulating isolation pattern **130**, and the gate dielectric layers **162** and the first conductive layer **171** may conformally extend along the side surface **130LS** of the insulating isolation pattern **130**. The second conductive layer **172** may be configured the same as or similar to the first conductive layer **171**.

[0073] FIG. **6** is a cross-sectional diagram illustrating a semiconductor device according to example embodiments, illustrating a region corresponding to FIG. **2B**.

[0074] Referring to FIG. **6**, a semiconductor device **100D** in an example embodiment may be configured the same as or similar to the example described with reference to FIGS. **1** to **5**, other than the configuration in which the ends of the first epitaxial layers **145**, opposing each other, are not in contact with the gate dielectric layers **162**. In the semiconductor device **100D**, at least a portion **130LS** of the side surface of the insulating isolation pattern **130** may extend parallel to the side surface **140S** on which the plurality of channel layers **140** oppose each other, and the ends **145e** of the first epitaxial layers **145** opposing each other may not be in direct contact with the gate dielectric layers **162**, and may be spaced apart from each other. In the semiconductor device **100D**, the end **145e** of the first epitaxial layers **145** may be in contact with the insulating isolation pattern **130**. The relationship between the first width **W1**, the second width **W2**, the third width **W3** and the fourth width **W4** of the insulating isolation pattern **130** in the semiconductor device **100D** may be the same as the relationship between the first width **W1**, the second width **W2**, the third width **W3** and the fourth width **W4**.

[0075] FIG. **7** is a cross-sectional diagram illustrating a semiconductor device according to example embodiments, illustrating a region corresponding to FIG. **2B**.

[0076] Referring to FIG. **7**, the semiconductor device **100E** in an example embodiment may be configured the same as or similar to the example described with reference to FIGS. **1** to **6** other than the configuration in which at least a portion of the side surface **130XS** of the insulating isolation pattern **130** has a convex shape on a level between the plurality of first channel layers **140A**. In the semiconductor device **100E**, a width of the insulating isolation pattern **130** in the second direction (e.g., the Y-axis direction) may not be uniform. The width of the insulating isolation pattern **130** in the second direction may be widest on a level between the plurality of first channel layers **140A** or the plurality of second channel layers **140B**, and may have a shape decreasing toward the end **145e** of the first epitaxial layers **145**. The side surface of the insulating isolation pattern **130** in the second direction may have a concave shape between the plurality of channel layers **140** disposed on the same level, and may have a convex shape between the first epitaxial layer **145**, disposed on a level adjacent thereto, and the lowermost first epitaxial layer **145** and the second epitaxial layer **146**, disposed on a lowest level among the first epitaxial layers **145**. The side surface of the insulating isolation pattern **130** in the second direction may have a serrated shape in which the concave shape and the convex shape described above are repeated.

[0077] On a level between the channel layers **140** spaced apart from each other in the vertical direction, the third width **W3** between the side surfaces **130CS** of the insulating isolation pattern **130**, opposing each other, may be larger than the first width **W1** of the portion of the insulating isolation pattern **130** between the ends **145e** of the first epitaxial layers **14**, opposing each other.

[0078] FIGS. **8A** to **8I** are diagrams illustrating processes of a method of manufacturing a semiconductor device according to example embodiments. FIGS. **8A** to **8I** illustrate an example embodiment of the method of manufacturing a semiconductor device **100** in FIGS. **1** to **2B**. FIGS. **8A** to **8I** illustrate cross-sectional surfaces taken along lines II-II' in FIG. **1**.

[0079] Referring to FIG. **8A**, sacrificial layers **120** and a plurality of channel layers **141**, **142**, and

143 may be alternately stacked on a substrate **101**.

[0080] The sacrificial layers **120** may be replaced with the gate dielectric layers **162** and the first and second gate electrodes **170A** and **170B** through a subsequent process, as illustrated in FIGS. **2A** and **2B**. Sacrificial layers **120** may be formed of a material having etch selectivity for the first to third channel layers **141**, **142**, and **143**, respectively. The first to third channel layers **141**, **142**, and **143** may include a material different from that of the sacrificial layers **120**. The sacrificial layers **120** and the first to third channel layers **141**, **142**, and **143** may include, for example, a semiconductor material including at least one of silicon (Si), silicon germanium (SiGe), and germanium (Ge), may include different materials, and may or may not include impurities. For example, the sacrificial layers **120** may include silicon germanium (SiGe), and the first to third channel layers **141**, **142**, and **143** may include silicon (Si).

[0081] The sacrificial layers **120** and the first to third channel layers **141**, **142**, and **143** may be formed by performing an epitaxial growth process from the substrate **101**. The number of channel layers **141**, **142**, and **143** alternately stacked with the sacrificial layers **120** may be varied in example embodiments. A thickness of the sacrificial layer **120U** disposed on an uppermost end among the plurality of sacrificial layers **120** may be greater than a thickness of the other sacrificial layers **120**, but an example embodiment thereof is not limited thereto.

[0082] Referring to FIG. **8B**, active structures may be formed by removing a portion of the sacrificial layers **120**, the plurality of channel layers **141**, **142**, and **143**, and the substrate **101**, and the device isolation layer **110** may be formed.

[0083] The active structures may include sacrificial layers **120** and a plurality of channel layers **141**, **142**, and **143** alternately stacked, and may further include an active region **105** protruding from the substrate **101** by removing a portion of the substrate **101**. The active structures may be formed in a line shape extending in the first direction (e.g., X-axis direction) and may be spaced apart from each other in the second direction (e.g., Y-direction). In example embodiments, when the active region **105** is divided into first and second active regions, the first and second active regions may include the same or different impurities, and the impurities may be doped in the substrate **101** before the sacrificial layers **120** and the plurality of channel layers **141**, **142**, and **143** are formed.

[0084] In the region from which a portion of the substrate **101** has been removed, the device isolation layer **110** may be formed by filling the insulating material and partially removing the insulating material such that the active region **105** may protrude. An upper surface of the device isolation layer **110** may be formed on a level lower than a level of an upper surface of the active region **105**.

[0085] Referring to FIG. **8C**, a first recess region **RC1** penetrating the sacrificial layers **120** and the channel layers **140** and extending in the third direction (e.g., Z-axis direction) may be formed on the active structure. The first recess region **RC1** may be formed by patterning the sacrificial layers **120** and the channel layers **140**. The first recess region **RC1** may be configured as a trench region extending in the first direction (e.g., X-axis direction) along the active structure. The first recess region **RC1** may extend into the substrate **101**, and the first recess region **RC1** may overlap at least a portion of each of the first region **R1** and the second region **R2** of the substrate **101** in the second direction (e.g., the Y-axis direction).

[0086] Referring to FIG. **8D**, a mask layer **135** may be disposed on both sides of the active structure, and in the first recess region **RC1**, a first epitaxial layer **145** may be formed on an internal side surface **140S** of each of the plurality of channel layers **140**, and a second epitaxial layer **146** may be formed on an internal side surface of the active structure.

[0087] The mask layer **135** may cover both side surfaces of the active structure and at least a portion of the upper surface of the active structure. The mask layer **135** may extend in the first direction (e.g., X-axis direction), parallel to the active structure. The mask layer **135** may not cover the first recess region **RC1**, and may cover both side surfaces of the active structure to prevent

silicon (Si) from epitaxially growing on both side surfaces in a subsequent process. Side surfaces in the first recess region RC1 on side surfaces of the channel layers **140** may be exposed from the mask layer **135**. The first epitaxial layer **145** and the second epitaxial layer **146** may be formed through an epitaxial growth process on one side surface of the channel layers **140** in the first recess region RC1, and may include the same material. For example, the first epitaxial layer **145** and the second epitaxial layer **146** may include silicon (Si). The first epitaxial layer **145** and the second epitaxial layer **146** may have a cross-sectional surface having a triangular shape, as illustrated, but an example embodiment thereof is not limited thereto, and The first epitaxial layer **145** and the second epitaxial layer **146** may have a cross-sectional surface having a trapezoidal shape (see FIG. 3). Also, after the first epitaxial layer **145** and the second epitaxial layer **146** are formed, a portion of etching or other removal processes may be performed. For example, by removing a portion of the end **145e** of the first epitaxial layer **145**, the semiconductor device (**100B**, see FIG. 4) of another example embodiment including the first epitaxial layer **145** having a rounded semicircular shape may be formed. A distance in the second direction (e.g., Y-axis direction) between the ends **145e** of the first epitaxial layers **145**, opposing each other, may be referred to as a first width W1. At least a portion of an upper surface of the second epitaxial layer **146** may have a concave shape toward the substrate **101** in a central portion adjacent to a boundary between the first region R1 and the second region R2. A width of the concave region of the second epitaxial layer **146** in the second direction may be referred to as a second width W2. A size of the first width W1 may be the same as a size of the second width W2, but an example embodiment thereof is not limited thereto. Depending on the degree to which the semiconductor material layers EP are formed, the first width W1 and the second width W2 may be determined, and the semiconductor device in the various example embodiments described above may be formed.

[0088] Referring to FIG. 8E, a preliminary insulating isolation pattern **130p** may be formed by filling the first recess region RC1 with an insulating material, and the mask layer (**135**, see FIG. 8D) may be removed.

[0089] The insulating material may include silicon oxide, silicon nitride, silicon oxynitride, or silicon carbide. The preliminary insulating isolation pattern **130p** may be formed on the active structure in the active region **105**, and may have a shape extending in the first direction (e.g., the X-axis direction) along the active structure. The preliminary insulating isolation pattern **130p** formed according to the process in which the insulating material fills the first recess region (RC1, see FIG. 8D) may cover the first epitaxial layer **145** and the second epitaxial layer **146**. An upper surface of the preliminary insulating isolation pattern **130p** may be coplanar with an upper surface of the uppermost sacrificial layer **120U**. In the subsequent process, by removing the mask layer **135** covering both side surfaces of the active structure and at least a portion of the upper surface, both side surfaces and upper surfaces of the active structure may be exposed, and at least a portion of the upper surface of the device isolation layer **110** may be exposed.

[0090] Referring to FIG. 8F, a sacrificial gate structure **200** and gate spacer layers **163** (see FIG. 2A) may be formed on the active structure.

[0091] The sacrificial gate structure **200** may be configured as a sacrificial structure formed in a region in which the gate dielectric layers **162** and the first and second gate electrodes **170A** and **170B** are disposed on the channel structures **140** through a subsequent process, as illustrated in FIGS. 2A and 2B. The sacrificial gate structure **200** may have a line shape intersecting the active structures and extending in one direction. The sacrificial gate structure **200** may extend, for example, in the Y-direction.

[0092] The sacrificial gate structure **200** may include first and second sacrificial gate layers **202** and **205** and a mask pattern layer **206** stacked in order. The first and second sacrificial gate layers **202** and **205** may be patterned using the mask pattern layer **206**. The first and second sacrificial gate layers **202** and **205** may be configured as an insulating layer and a conductive layer, respectively, but an example embodiment thereof is not limited thereto, and the first and second

sacrificial gate layers **202** and **205** may be configured as an integrated layer. For example, the first sacrificial gate layer **202** may include silicon oxide, and the second sacrificial gate layer **205** may include polysilicon. The mask pattern layer **206** may include silicon oxide and/or silicon nitride. [0093] The gate spacer layers (**163**, see FIG. 2A) may be formed in order on both sidewalls of the sacrificial gate structure **200**. The gate spacer layers **163** may be formed of a low dielectric constant material and each of the gate spacer layers **163** may include at least one of SiO, SiN, SiCN, SiOC, SiON, and SiOCN, for example.

[0094] First, recess regions may be formed by removing a portion of the exposed sacrificial layers **120** and plurality of channel layers **141**, **142**, and **143** using the sacrificial gate structure **200** and gate spacer layers **163** as a mask. Accordingly, the plurality of channel layers **141**, **142**, and **143** may form channel structures **140** having a limited length in the first direction (e.g., X-axis direction).

[0095] Thereafter, source/drain regions **150** may be formed by growing from the side surfaces of the active regions **105** and the channel structures **140**, for example, by a selective epitaxial process. The source/drain regions (**150**, see FIGS. 2A and 2C) may include impurities due to in-situ doping and may include a plurality of layers having different doping elements and/or doping concentrations.

[0096] Referring to FIG. 8G, the interlayer insulating layer IL1 may be formed, and the sacrificial layers **120** and the sacrificial gate structure **200** may be removed.

[0097] The interlayer insulating layer **190** may be formed by forming an insulating film covering the sacrificial gate structure **200** and the source/drain regions **150** and performing a planarization process.

[0098] The sacrificial layers **120** and the sacrificial gate structure **200** may be selectively removed with respect to the gate spacer layers **163**, the interlayer insulating layer (IL1, see FIG. 2A), and the channel structures **140**. First, an upper gap region (not illustrated) may be formed by removing the sacrificial gate structure **200**, and lower gap regions (not illustrated) may be formed by removing the sacrificial layers **120** exposed through the upper gap region. For example, when the sacrificial layers **120** include silicon germanium (SiGe) and the channel structures **140** include silicon (Si), the sacrificial layers **120** may be selectively removed by performing a wet etching process. Depending on the process of removing the sacrificial layers **120**, the side surface **130ES** of the preliminary insulating isolation pattern **130p** in contact with the sacrificial layers **120** may be exposed from the plurality of channel layers **141**, **142**, and **143**.

[0099] Referring to FIG. 8H, a second recess region RC2 and an insulating isolation pattern **130** may be formed by removing a portion of the preliminary insulating isolation pattern **130p**.

[0100] A portion of the exposed side surface (**130ES**, see FIG. 8G) of the preliminary insulating isolation pattern **130p** may be removed by a dry etching process. When removed by a dry etching process, as illustrated in FIG. 8H, a portion **130CS** of the insulating isolation pattern **130** on the side surface may have a concave shape, but an example embodiment thereof is not limited thereto. In an example embodiment, when an ALE (Atomic Layer Etching) process is performed, the etching process may be performed parallel to the exposed surface **130ES** of the preliminary insulating isolation pattern **130p**, and the side surface of the insulating isolation pattern **130** may have a linearly extending shape (**130LS**, see FIG. 5). When the ALE process is performed, the etching speed may not be necessarily constant in the surface direction, and etching may be performed faster in the central portion. In this case, the semiconductor device (**100D**, see FIG. 6) in which the etching process has been performed less on an outer region of the exposed surface **130ES** of the preliminary insulating isolation pattern **130p** may be formed. In another example embodiment, the etching process may be performed faster along the surface on the upper surface and the lower surface of the channel layers **140**, and in this case, the semiconductor (**100E**, see FIG. 7) in which the etching process is further performed on the outer region of the exposed surface **130ES** of the preliminary insulating isolation pattern **130p** may be formed. By the process of

forming the second recess region RC2 by removing a portion of the preliminary insulating isolation pattern **130p**, a third width W3 between the side surfaces **130CS** of the insulating isolation pattern **130** opposing each other may be determined on a level between the channel layers **140** vertically spaced apart from each other. The fourth width W4 of the insulating isolation pattern **130** on a level higher than a level of the uppermost channel layer **143** among the plurality of channel layers **140** may be greater than the third width W3, but an example embodiment thereof is not limited thereto. By the process of forming the second recess region RC2, the semiconductor device of various example embodiments described above may be formed.

[0101] Referring to FIG. 8I, gate dielectric layers **162** may be formed, a first conductive layer **171** and a second conductive layer **172** may be formed, and a preliminary third conductive layer **173p** may be formed.

[0102] The gate dielectric layers **162** may be formed to conformally cover a side surface, a lower surface and an upper surface of the plurality of channel layers **140** and a side surface of the insulating isolation pattern **130** on the first and second regions R1 and R2. The gate dielectric layers **162** may be formed to cover both the first region R1 and the second region R2 of the substrate **101**. The first conductive layer **171** may be formed on the first region R1, and the second conductive layer **172** may be deposited on the second region R2. Using a mask pattern (not illustrated), the first conductive layer **171** and the second conductive layer **172** may be formed on the insulating isolation pattern **130** such that one side surface may be in contact with. The first conductive layer **171** and the second conductive layer **172** may control a work function and may include different metals, but an example embodiment thereof is not limited thereto. A preliminary third conductive layer **173p** may be formed on the first conductive layer **171** and the second conductive layer **172**. The preliminary third conductive layer **173p** may fill a region between the plurality of first channel layers **140A** and the plurality of second channel layers **140B**, and may overlap the first epitaxial layers **145** in the third direction (e.g., the Z-axis direction). The preliminary third conductive layer **173p** may extend further than the side surface **140S** of the plurality of channel layers **140** opposing each other.

[0103] Thereafter, referring to FIGS. 2A to 2C, in a subsequent process, a third conductive layer **173**, gate capping layers **166**, and contact plugs **180** may be formed, thereby forming the semiconductor device **100** in the example embodiment.

[0104] The third conductive layer **173** may be formed by further performing a planarization process after a process of depositing the preliminary third conductive layer (**173p**, see FIG. 8I). The third conductive layer **173** may completely fill an upper gap region (not illustrated). By the planarization process, an upper surface of the insulating isolation pattern **130** may be exposed, and an upper surface of the insulating isolation pattern **130** and an upper surface of the third conductive layer **173** may be coplanar with each other. In some example embodiments, the third conductive layer **173** may include a plurality of conductive layers. Accordingly, consequently, the first and second gate electrodes **170A** and **170B** may be formed.

[0105] The gate capping layers **166** may be formed by partially removing the first and second gate electrodes **170A** and **170B**, the gate dielectric layers **162**, and the gate spacer layers **163**, filling the removed regions with an insulating material, and performing a planarization process. A relative thickness of the gate capping layers **166** and the shape of lower surfaces thereof may be varied in example embodiments.

[0106] Thereafter, the second interlayer insulating layer IL2 may be further formed, and contact plugs **180** connected to the source/drain regions **150** may be formed. Accordingly, the semiconductor device **100** in FIGS. 1 to 2C may be manufactured.

[0107] According to the aforementioned example embodiments, by including channel layers of which a portion of a side surface is extended and an insulating pillar of which a portion of a side surface is removed, a semiconductor device having improved electrical properties and reliability may be provided.

[0108] While the example embodiments have been illustrated and described above, it will be configured as apparent to those skilled in the art that modifications and variations may be made without departing from the scope in the example embodiment as defined by the appended claims.

Claims

1. A semiconductor device, comprising: a substrate including a first region and a second region, the substrate including an active region extending in a first direction; a device isolation layer defining the active region, the device isolation layer being on the substrate; a first gate electrode and a second gate electrode extending in a second direction, the second direction intersecting the active region, the first gate electrode and the second gate electrode being spaced apart from each other; a plurality of first channel layers spaced apart from each other in a third direction, the third direction being perpendicular to an upper surface of the substrate, and the plurality of first channel layers being surrounded by the first gate electrode on the first region; a plurality of second channel layers spaced apart from each other in the third direction, the plurality of second channel layers being surrounded by the second gate electrode on the second region; an insulating isolation pattern extending in the first direction between the first gate electrode and the second gate electrode and between the plurality of first channel layers and the plurality of second channel layers, the insulating isolation pattern being on the substrate; gate dielectric layers between the plurality of first channel layers and the first gate electrode, and the gate dielectric layers between the plurality of second channel layers and the second gate electrode; and epitaxial layers on side surfaces of the plurality of first channel layers and the plurality of second channel layers that respectively oppose each other, wherein a portion of each of the epitaxial layers overlap the insulating isolation pattern, the gate dielectric layers, the first gate electrode, and the second gate electrode in the third direction.
2. The semiconductor device of claim 1, wherein the first gate electrode includes a first conductive layer conformally extending along an upper surface, a side surface, and a lower surface of each of the plurality of first channel layers, respectively, and wherein the second gate electrode includes a second conductive layer conformally extending along an upper surface, a side surface, and a lower surface of each of the plurality of second channel layers, respectively.
3. The semiconductor device of claim 2, wherein the first conductive layer and the second conductive layer include different metal materials.
4. The semiconductor device of claim 2, wherein the first gate electrode further includes a third conductive layer on the first conductive layer and between the plurality of first channel layers in the third direction.
5. The semiconductor device of claim 4, wherein the third conductive layer overlaps at least a portion of a corresponding one of the epitaxial layers in the third direction.
6. The semiconductor device of claim 1, wherein the insulating isolation pattern includes one of silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, or a combination thereof.
7. The semiconductor device of claim 1, wherein the insulating isolation pattern is between the first gate electrode and the second gate electrode.
8. The semiconductor device of claim 1, wherein, the epitaxial layers include a first one of the epitaxial layers on a side surface of a corresponding one of the plurality of first channel layers and a second one of the epitaxial layers on a side surface of a corresponding one of the plurality of second channel layers, the first one of the epitaxial layers and the second one of the epitaxial layers are disposed at a same level and spaced apart from each other with the insulating isolation pattern therebetween.
9. The semiconductor device of claim 1, wherein each of the epitaxial layers has a width decreasing in the third direction as the epitaxial layers extend toward the insulating isolation pattern.
10. The semiconductor device of claim 9, wherein a cross-sectional shape of each of the epitaxial

layers is one of a triangular shape, a quadrangular shape, or a semicircular shape.

11. The semiconductor device of claim 1, further comprising: a gate capping layer on an upper surface of the first gate electrode, an upper surface of the second gate electrode, and an upper surface of the insulating isolation pattern.

12. The semiconductor device of claim 1, wherein the epitaxial layers each include a same material as a material of an adjacent one of the plurality of first channel layers or a material of an adjacent one the plurality of second channel layers adjacent to each other.

13. The semiconductor device of claim 1, wherein a portion of a side surface of the insulating isolation pattern extends in the third direction along ends of the epitaxial layers in the second direction.

14. The semiconductor device of claim 13, wherein ends of the epitaxial layers, opposing each other are not in contact with the gate dielectric layers.

15. The semiconductor device of claim 1, wherein a portion of a side surface of the insulating isolation pattern has a convex shape toward an external side on a level between the epitaxial layers on different levels.

16. A semiconductor device, comprising: a substrate including an active region extending in a first direction; a first gate electrode and a second gate electrode extending in a second direction, the second direction intersecting the active region, the first gate electrode and the second gate electrode being spaced apart from each other; a plurality of first channel layers spaced apart from each other in a third direction, the third direction perpendicular to an upper surface of the substrate, and the plurality of first channel layers being surrounded by the first gate electrode; a plurality of second channel layers spaced apart from each other in the third direction, the plurality of second channel layers being surrounded by the second gate electrode; an insulating isolation pattern between the plurality of first channel layers and the plurality of second channel layers, the insulating isolation pattern extending to a level lower than a level of an upper surface of the active region; first epitaxial layers on side surfaces of the plurality of first channel layers and the plurality of second channel layers, opposing each other, the first epitaxial layers being in contact with the insulating isolation pattern in the active region; and a second epitaxial layer in the active region of the substrate and having an upper surface in contact with the insulating isolation pattern.

17. The semiconductor device of claim 16, wherein the first epitaxial layers and the second epitaxial layer include a same material.

18. The semiconductor device of claim 16, wherein a lower surface of the second epitaxial layer is lower than a level of an upper surface of the active region.

19. A semiconductor device, comprising: a substrate including a first region and a second region, the substrate including an active region extending in a first direction; a first gate electrode and a second gate electrode extending in a second direction, the second direction intersecting the active region, the first gate electrode including a first conductive layer, the second gate electrode including a second conductive layer, the first gate electrode and the second gate electrode each including a third conductive layer, the first conductive layer and the third conductive layer of the first gate electrode being stacked in order, and the second conductive layer and the third conductive layer of the second gate electrode being stacked in order; a plurality of channel layers stacked in a third direction on the active region, the third direction being perpendicular to an upper surface of the substrate, and the plurality of channel layers overlapping the third conductive layer of the first gate electrode in the third direction; a gate dielectric layer covering a portion of each of the plurality of channel layers; an insulating isolation pattern between the first gate electrode and the second gate electrode; and a semiconductor material layer disposed on a side surface of at least one of the plurality of channel layers, wherein the first conductive layer and the second conductive layer include different metal materials.

20. The semiconductor device of claim 19, wherein the gate dielectric layer is between the first gate

electrode and the insulating isolation pattern and between the second gate electrode and the insulating isolation pattern.
